ORGANIC CHEMISTRY FRONTIERS

RESEARCH ARTICLE

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Cite this: *Org. Chem. Front.*, 2025, **12**, 185

Regio- and stereoselective azidation of activated *N*-allenamides: an entry to α, β, γ and δ-amido-azides†

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Received 25th September 2024,
Accepted 31st October 2024

DOI: 10.1039/d4qo01802h

rsc.li/frontiers-organic

A totally controlled regiodivergent azidation of activated *N*-allenamides is presented. Using $TMSN_3$/TBAF, β-azidation of *N*-allenamides occurs exclusively, yielding vinyl azides. Conversely, employing a TFA/$TMSN_3$ mixture results solely in the formation of γ-azides. A subsequent [1,3] azide shift of the latter with DBU produces α-amido vinyl azides. Additionally, δ-difluorinated azides featuring an ynamide are selectively synthesized from ene-ynamides. The practical applicability of these transformations is demonstrated through the formation of cyanide derivatives, trifluoromethyl ketones and primary enamines.

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Introduction

Organic azides are versatile synthetic intermediates.[1] As effective ammonia surrogates,[2] they serve as keystones for the synthesis of N-heterocycles,[3] and offer access to essential bioactive molecules.[4] Among them, the unique features of vinyl-azides have turned these compounds into critical building blocks for the construction of N-heterocycles.[5] The remarkable properties of the azide group connected to an alkene moiety[6] allows this functional group to act as an electrophile,[7] an enamine-type nucleophile[8] or a radical acceptor.[9] The multifaced reactivity of this highly versatile synthons have generated a great inspiring variety of intermediates such as iminodiazonium ions, nitrilium ions,[10] iminyl radicals[11] and metal enaminyl radicals[12] upon heating-, metal-or light-induced processes.[5d] Classical procedures to prepare vinyl azides involve base-promoted elimination of haloazidoalkanes[13] and hydroazidation of alkynes.[14] The incorporation of a trifluoromethyl group into biologically active compounds frequently enhances their physicochemical properties, including lipophilicity and metabolic stability.[15] Consequently, molecules that contain both a fluorine group and an azide moiety serve as highly effective building blocks in life science and agrochemistry.[16] Following our interest in the preparation of nitrogen-containing building blocks, we envisioned that *N*-allenamides would be ideal candidates to generate vinyl azides. Copper-catalyzed carboazidation of allenes allowed the formation of 3-(azidomethyl)-2*H*-azirines (Scheme 1a).[17] Allyl azides have been obtained either from allenes, (Scheme 1b and c)[18] or *N*-allenamides through metal-catalyzed processes (Scheme 1d).[19] To obtain trifluoromethylated azides, another alternative consists in performing a trifluoromethyl azidation.[20] Likewise a copper-catalyzed trifluoromethylazidation of allenes predominantly led to CF_3 containing allyl azides (Scheme 1e).[21]

Results and discussion

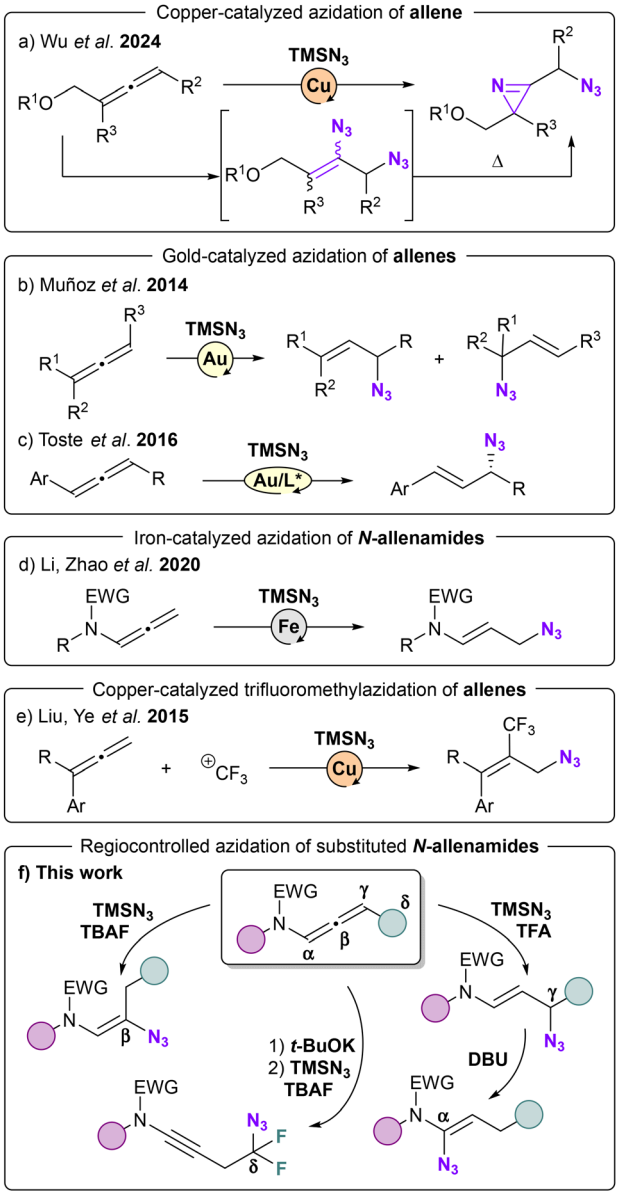
We initially investigated the azidation of CF_3-substituted *N*-allenamides. Trifluoromethylated *N*-allenamides were obtained by treatment of terminal ynamides with trifluoromethylated diazomethane according to our previously developed strategy.[22] Based on our previous findings regarding the addition of a nucleophile on activated *N*-allenamides,[23] we anticipated that a nucleophilic azide would add on the central sp carbon of CF_3-*N*-allenamides to form vinyl azides equipped with a tosylsulfonamide and a trifluoromethylated moiety. When we treated CF_3-substituted *N*-allenamide **1a** with sodium azide and 15-Crown-5, the targeted vinyl azide **2a** was obtained although with a moderate yield of 28% (Table 1, entry 1). Modification of the solvent did not improve the result (Table 1, entries 2–5). The reaction failed to proceed without crown-ether (Table 1, entry 6). However, using tetrabutylazide as an azide source revealed to be more efficient for this transformation (Table 1, entry 7). Finally, a mixture of $TMSN_3$ and TBAF proved to be the optimal choice, affording the corresponding vinyl azide **2a** in 84% yield (Table 1, entry 8). X-ray analyses of **2c** confirmed the structure of the β-amido-vinyl

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†Electronic supplementary information (ESI) available: All experimental data and detailed procedures, including computational details. CCDC 2384604, 2384608 and 240703. For ESI and crystallographic data in CIF or other electronic format see DOI: https://doi.org/10.1039/d4qo01802h

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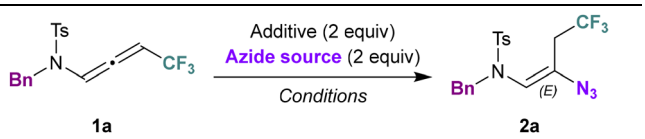
Scheme 1 Azidation of allene derivatives.

azide obtained (CCDC 2384608 contains the ESI† for the structure).[24]

With the optimized reaction conditions in hand, the substrate scope was investigated (Scheme 2). First, we examined different side chains on the nitrogen atom. Aryl substituents (**2a**, **2b**, **2c**) were tolerated for this transformation. Pleasingly, functionalized side chains suitable for late-stage transformations could be accommodated. In particular, unsaturated side chains (**2d**, **2e**), tri-fluorinated side chains (**2f**) imides (**2g**) and cycloalkanones (**2h**) were successfully adapted. Mesylated as well as cyclic sulfonamides and oxazolidinone were appropriate substrates and provided the corresponding alkenyl azides with a total site and stereoselectivity (**2i–k**).

The process has been applied to *N*-allenamides equipped with an ester function to provide the corresponding vinyl

Table 1 Optimization of the azidation process on the central sp carbon of *N*-allenamides



1a 2a

Entry[a]	Azide source	Additive	Solvent	Temp. (°C)	Yield **2a**[b] (%)
1	NaN_3	15-Crown-5	CH_2Cl_2	23	28
2	NaN_3	15-Crown-5	CH_3CN	0	30
3	NaN_3	15-Crown-5	THF	23	Traces
4	NaN_3	15-Crown-5	PhMe	23	Traces
5	NaN_3	15-Crown-5	DMF	23	Traces
6	NaN_3	—	CH_2Cl_2	23	0[c]
7	$TBAN_3$	—	THF	0	60
8	**$TMSN_3$**	**TBAF[d]**	**THF**	**0**	**84**
9	$TMSN_3$	TBAF[d]	CH_3CN	0	79

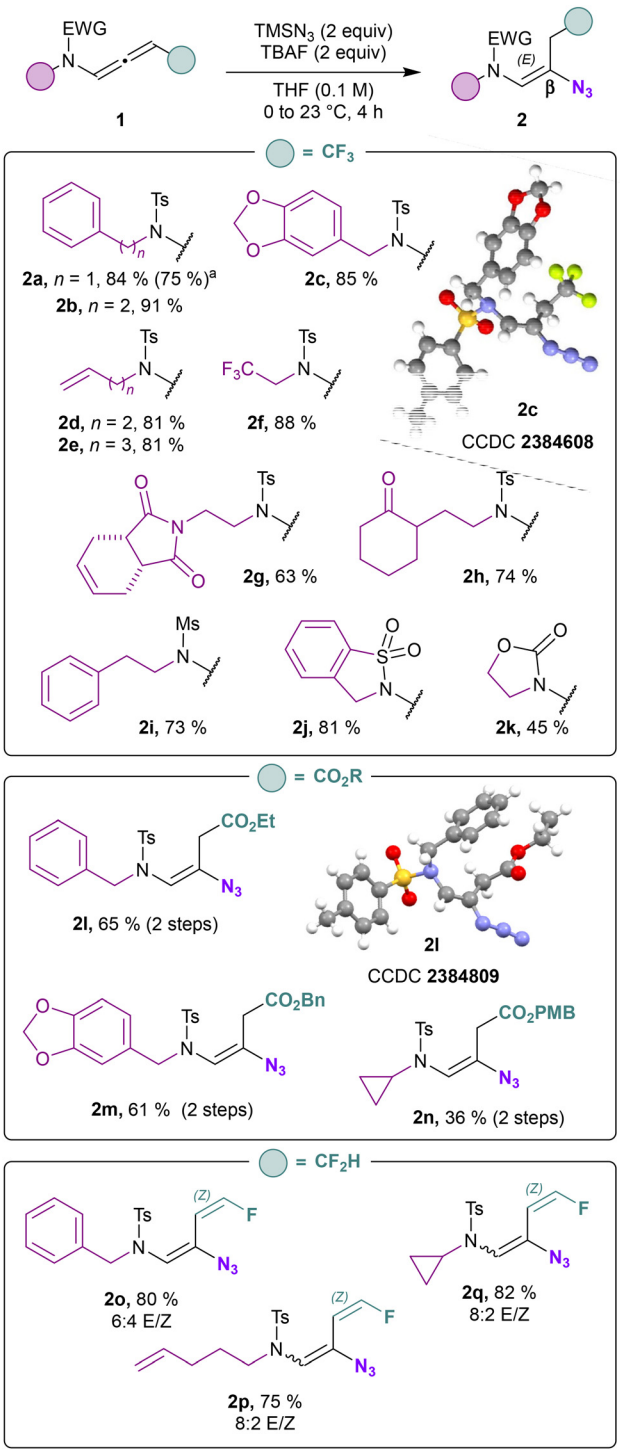
[a] Reaction conditions: **1a** (0.1 mmol), azide reagent (0.2 mmol), additive (0.2 mmol), in solvent (0.1 M) under N_2 atmosphere. [b] Isolated yields. [c] Starting material was fully recovered. [d] 1 M solution in THF.

azides (**2l–n**) in similar yields. Due to instability of ester-substituted *N*-allenamides, the latter has not been isolated before hydroazidation and the alkenyl azides are obtained sequentially from the corresponding ynamides. X-ray analyses of **2l** confirmed the structure of the vinyl azides **2l–n** bearing an ester function (CCDC 2384809 contains the ESI† for the structure).[25] For difluorinated *N*-allenamides, an additional β-fluoride elimination takes place, resulting in the formation of an *E*/*Z* mixture of vinyl azides (**2o–q**) equipped with a *Z*-configurated monofluorinated alkene.[26] This outcome is significant, since the monofluoroalkene motif represents an appealing target for drug design because it serves as a peptide and enol bond isostere.[27] It is important to notice that the hydroazidation reaction on the central sp carbon is totally regio- and stereo-selective for CF_3- and CO_2R- substituted *N*-allenamides. Another question that emerged from our study is whether it would be possible to introduce the azide moiety on the α-position of the nitrogen. Delocalization of the nitrogen lone pair toward the allenic moiety confers a dual reactivity to these entities leading to proximal or distal adducts in a regio- and stereo-controlled manner.[28] Halogenofunctionalization,[29] metal-catalyzed hydrofunctionalizations[30] as well as acid-promoted hydrocarboxylation[31] favor this mode of activation of *N*-allenamides. When we exposed our CF_3-substituted *N*-allenamides to a mixture of $TMSN_3$ and TFA, we wondered whether a 1,2-addition would occur on the iminium ion to generate α-azido-enamides. In this case, allylic azide **3a** was exclusively observed (Scheme 3). The combination of TFA and $TMSN_3$ likely generates HN_3 *in situ* which is presumably the active reagent.[32] Considering literature precedents,[18,31] the observed results could have been anticipated although, solely terminal *N*-allenamides have been investigated.

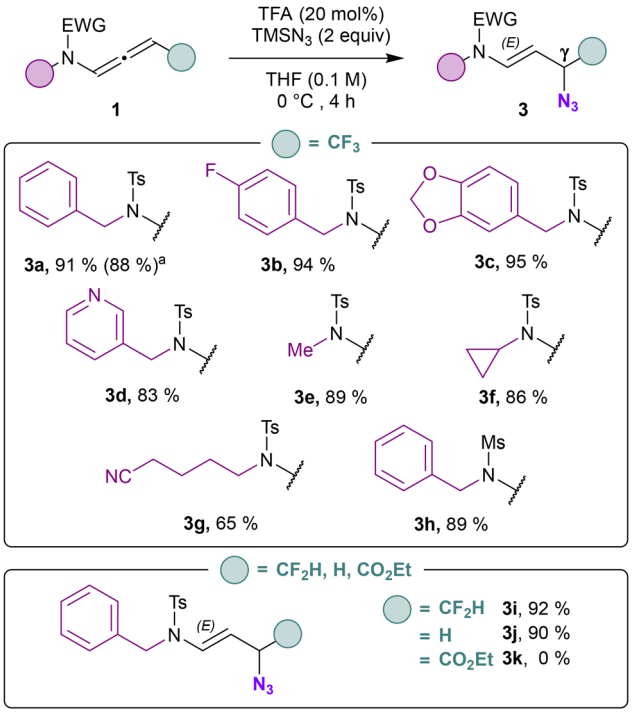
The distal hydroazidation was investigated with respect to the side chain on the nitrogen atom (Scheme 3). Benzylic

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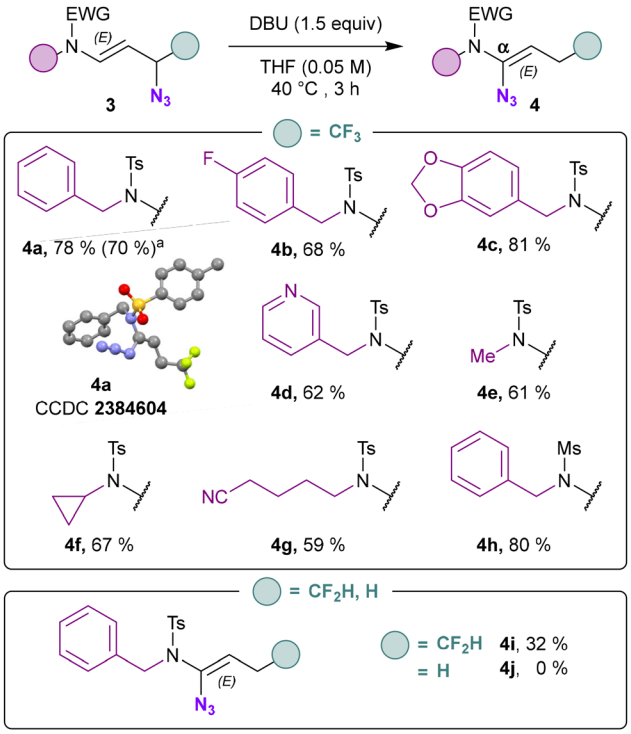
Scheme 2 Scope for the synthesis of β-amido azides. [a]This reaction was carried out on a 1 mmol scale.



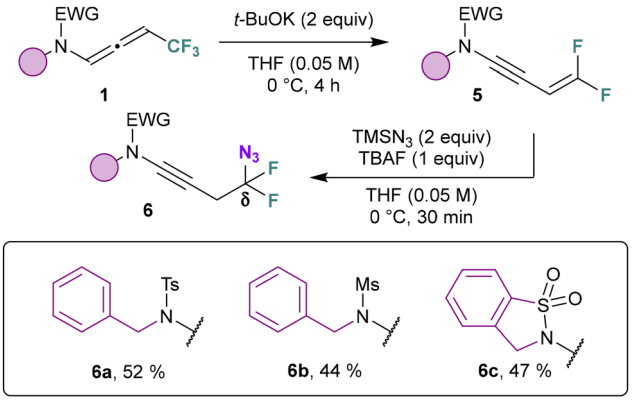
Scheme 3 Scope for the synthesis of γ-amido azides. [a]This reaction was carried out on a 1 mmol scale.

derivatives bearing either an electron-withdrawing (**3b**) and an electron-donating (**3c**) substituent were suitable compounds. Pyridine derivative provided the γ-amido azide **3d** in 83% yield. Linear alkyls (**3e**) and cyclic alkyls (**3f**) could be adapted while a nitrile-containing side led to the corresponding azide **3g** with an erosion of the yield compared to other substrates. Mesylated derivative also led to the formation of an γ-amido azide **3h** with 89% yield. Of note, difluorinated *N*-allenamide provided allylic azide **3i** in excellent yield (92%). Terminal *N*-allenamides also led to the corresponding allyl azide **3j** with $TMSN_3$ and Brønsted acid (TFA). The ester derivative proved to be unstable under these conditions and only degradation was observed (**3k**). When we treated the γ-azido derivative **3a** with a base, surprisingly, a [1,3] azide shift took place (Scheme 4). Only DBU provided exclusively *E*-configurated vinyl azide **4a**.[33] X-ray analyses of **4a** confirmed the structure of the α-amido-vinyl azide obtained (CCDC 2384604 contains the ESI† for the structure).[34] TMG provided a similar result albeit with a much lower yield of 43%. *t*-BuOK, NaH, led partly to hydrolysis of the allyl azide whereas with DMAP no conversion was observed. While vinyl azides have found extensive utility in organic synthesis, in contrast, allylic azides have often been avoided due to the propensity of the Winstein rearrangement. The latter leads to the presence of mixture of isomers through allylic azide equilibrium, thereby complicating the use of these intermediates in synthesis. The proximity of the heteroatom seems to alleviate those difficulties allowing to orientate the position of the azido moiety in α position.[32] With the optimized reaction conditions in hand, we evaluated the scope of this transformation. Benzylic derivatives (**4a–c**), heterocyclic derivatives (**4d**), linear alkyls (**4e**), cyclic alkyls (**4f**), functionalized side chains (**4g**), as well as mesylated derivatives (**4h**) underwent this transformation with yields ranging from 59 to 81% with total regio- and stereo-selectivity. CF_2H-substituted *N*-allenamides led to fluorinated vinyl-azides **4i** albeit with a

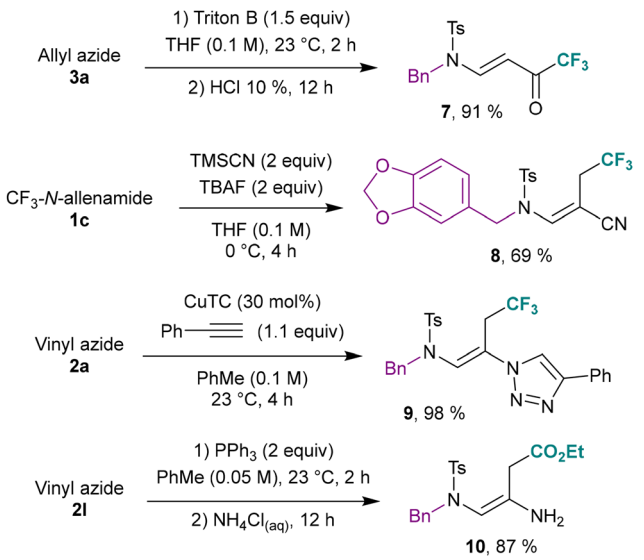
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Scheme 4 Scope for the synthesis of α-amido azides. [a]This reaction was carried out on a 1 mmol scale.



Scheme 5 Scope for the synthesis of δ-amido azides.



Scheme 6 Extensions and post-functionalization.

moderate yield (32%) probably due to concomitant elimination of a fluoride before [1,3]-azide shift. Terminal *N*-allenamides constitute the limit of this transformation, in this case the [1,3]-azide shift did not take place (**4j**). Having found that *gem*-difluorinated ene-ynamides could be obtained *via* deprotonation of trifluoromethylated *N*-allenamides and *δ* extrusion of fluorine,[35] we wondered whether the azide moiety could be installed *via* activation of the ynamide moiety in those substrates. When we treated ene-ynamide **5a** with $TMSN_3$, TBAF, surprisingly, the azide moiety was incorporated at δ-position in compound **6a** while the ynamide moiety remained unchanged (Scheme 5). In contrast to simple fluorinated organic azides, α-fluorinated azidoalkanes have been less studied, despite their much higher stability compared to their non-fluorinated counterparts.[36] Benzyl, mesyl as well as cyclic sulfonamides could be adapted and the corresponding homopropagylic azide (**6a–c**) could be isolated with 52, 44 and 47% yields respectively.

To further highlight these transformations, it is noteworthy that allyl-azide **3a** could be efficiently hydrolyzed providing the corresponding trifluorinated ketone **7** using Triton B in excellent yield (91%). These latter compounds are valuable synthetic intermediates for constructing fluorinated pharmacons, which typically require harsh conditions for their synthesis.[37] It should be noted that only benzylic azides have been reported as suitable substrates for hydrolysis.[38] Nitriles were also effective as nucleophiles and addition on the central sp carbon of a CF_3-*N*-allenamide provided exclusively the *E*-configurated vinyl cyanide **8**. Vinyl azide **2a** also proved to be an excellent partner to ethynyl benzene for click chemistry, under copper catalysis affording 1,4-triazole **9** almost quantitatively.[14b] Furthermore, primary enamines **10** could be isolated by reduction of vinyl azides leading to interesting building blocks bearing a free amine function (Scheme 6).

Conclusion

A totally regio- and stereoselective azidation of activated *N*-allenamides has been developed. Azidation of central sp carbon takes place with CF_3 and ester substituted *N*-allenamides. Difluorinated *N*-allenamides produced an *E*/*Z* mixture of vinyl azides bearing a monofluorinated alkene with good overall yields. γ-Azidation was tolerated for terminal as well as CF_3-substituted *N*-allenamides. A [1,3] azide shift with DBU provided α-substituted amido-vinyl azides starting from fluorinated *N*-allenamides. Hydroazidation *via* an ene-ynamide

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took place exclusively on the alkenyl *gem*-difluorinated moiety delivering difluorinated products while leaving the ynamide part intact. Finally, it was demonstrated that this transformation could be adapted to form vinyl cyanids. More importantly, trifluoromethylated ketones and primary enamines, valuable building blocks for drug design could be formed easily.

Author contributions

L.M, D.S, and M.H conceived and designed the experiments. L.M directed the project. D.S and M.H performed the experiments. L.M wrote the paper. L.M, D.S, and M.H discussed the results and commented on the manuscript.

Data availability

All experimental data and detailed procedures, including computational details, are available in the ESI.†

Conflicts of interest

There are no conflicts to declare.

Acknowledgements

Support for this work was provided by CNRS and Université de Strasbourg. D. S. and M. H. thanks M.R.T. for a research fellowship.

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